

Shape of data



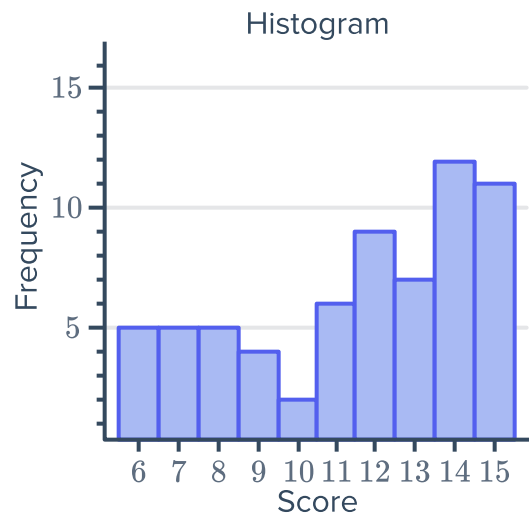
1 Describe the shape of the data in the following graphs:

a

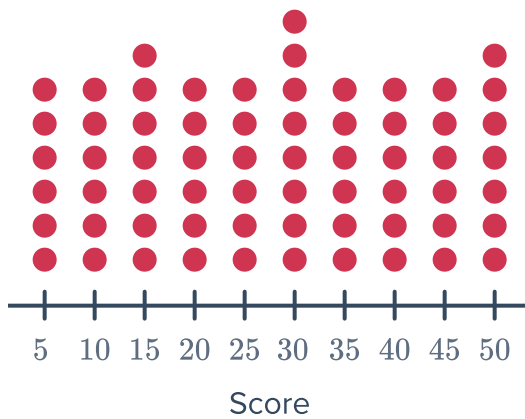
Leaf	
1	6 7 7
2	2 2 2 2 3 3 3
3	3 3 3 6 6 6 7 7 7 7 7
4	4 4 4 4 4 4
5	7 7

Key: 2|3 = 23

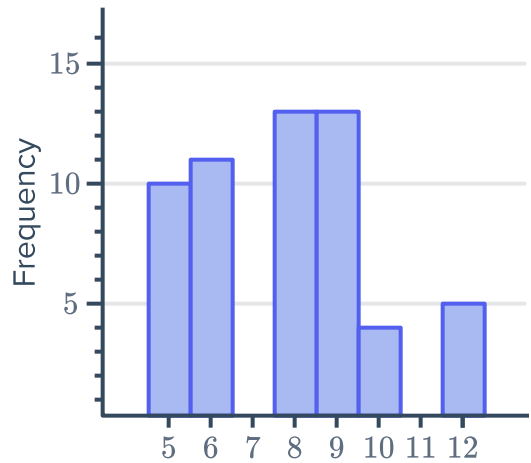
b



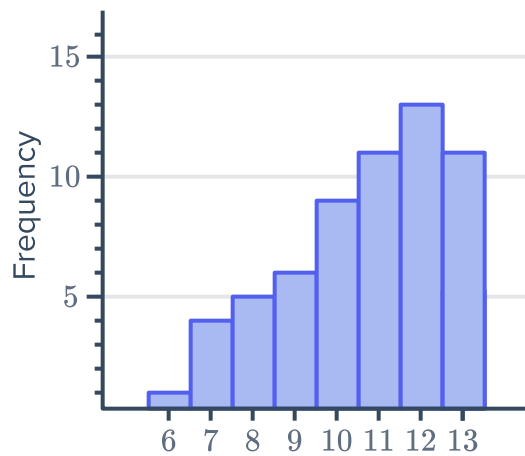
c



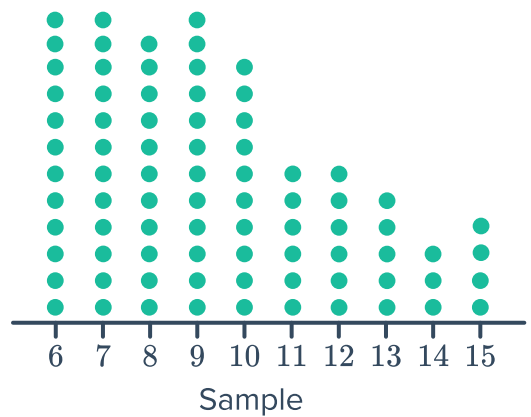
d



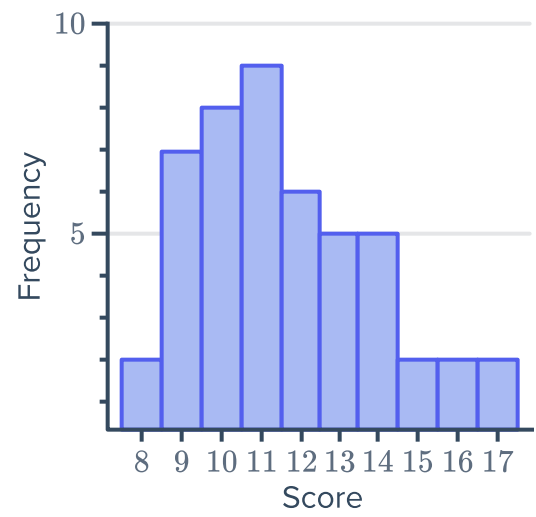
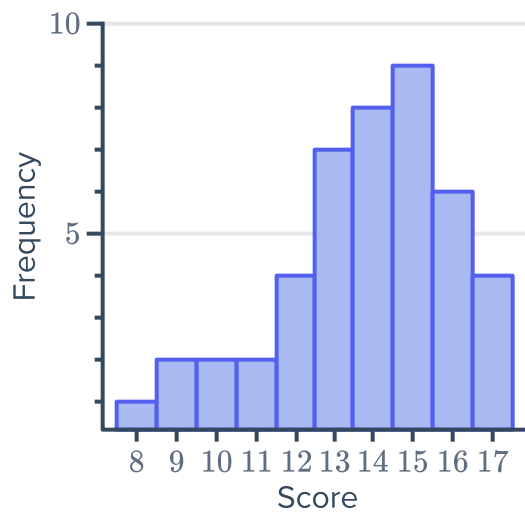
e



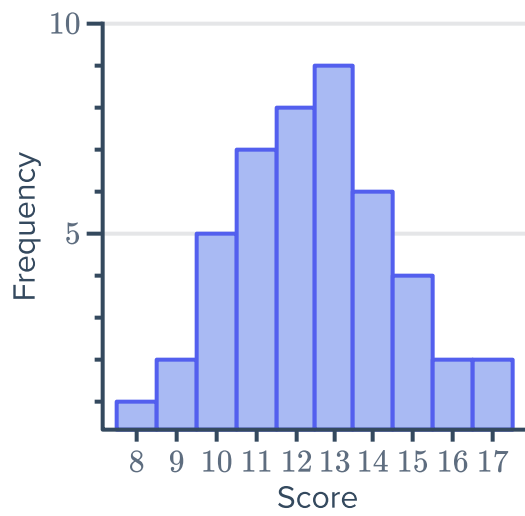
f



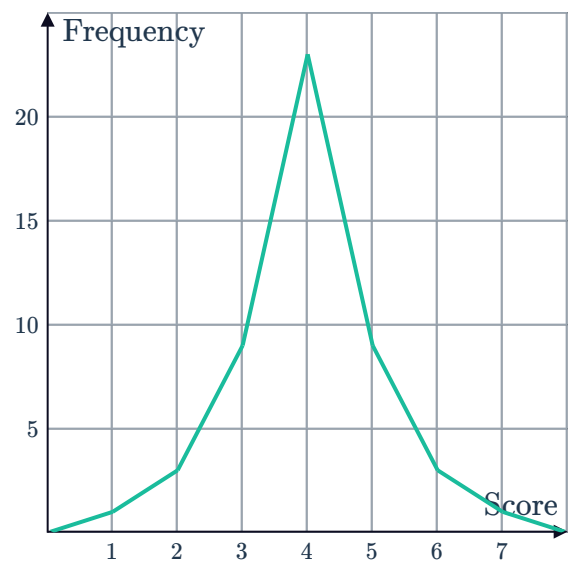
g



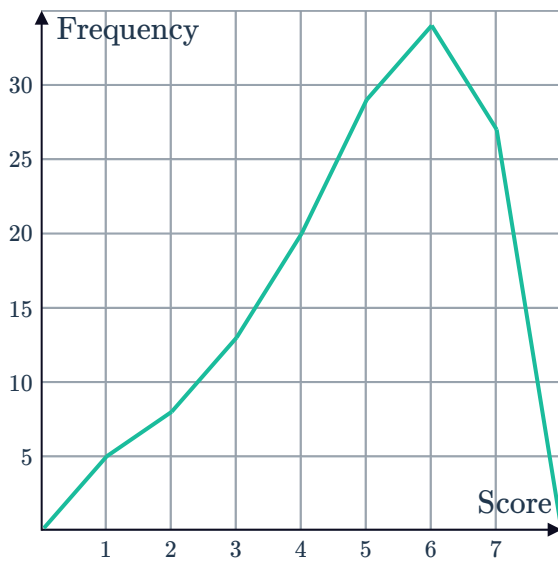
i



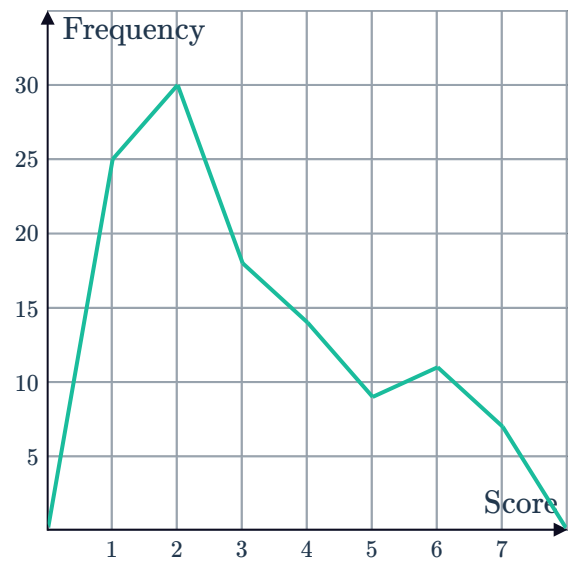
j



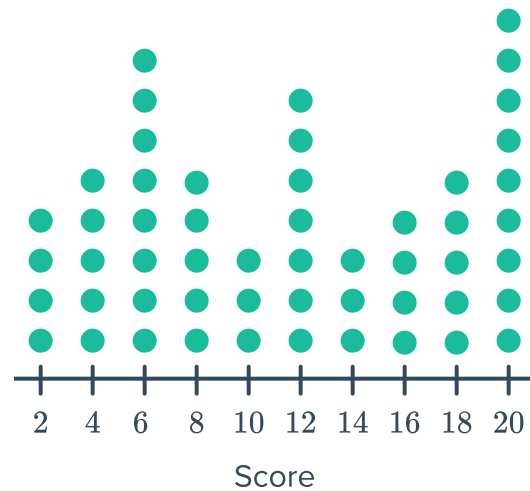
k



l



- 2 How would you describe the modality of the following dot plot? Explain your answer.

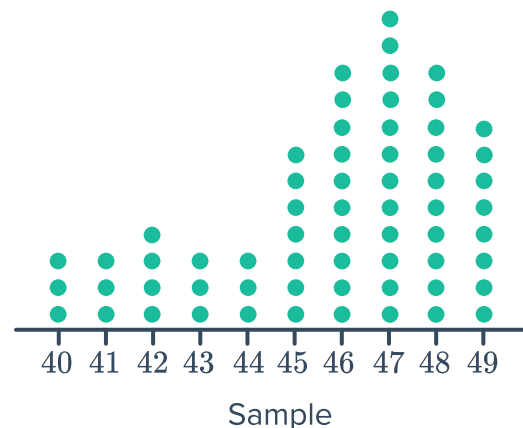


- 3 The table shows the number of crime novels in a bookshop for different price ranges rounded off to the nearest \$5:
- Graph this data in a histogram.
 - Describe the shape of the distribution of the data.

Price of crime novel	Frequency
\$5	5
\$10	10
\$15	17
\$20	8
\$25	17
\$30	10
\$35	5

Clusters and outliers

- 4 Temperatures were recorded over a period of time and presented as a dot plot:
- Are there any outliers?
 - Is there any clustering of data? If so, in what interval?
 - What is the modal temperature?
 - Describe the shape of the distribution.





- 5 The number of hours worked per week by a group of people is represented in the following stem and leaf plot:

- a State the value of any outliers.
- b Is there any clustering of data? If so, in what interval?
- c State the mode.

Leaf	
0	2
1	
2	0 3 6 6
3	1 4 5 6 6 7
4	0 4 6 7 9
5	0

Key: $2|3 = 23$

- 6 Consider the stem plot given:

- a State the value of any outliers.
- b Is there any clustering of data? If so, in what interval?
- c State the mode.
- d Describe the shape of the distribution.

Leaf	
0	5
1	7 8
2	0 8
3	1 3 3 7 8 9
4	1 3 5 8 8 8
5	
6	
7	
8	
9	2

Key: $2|3 = 23$

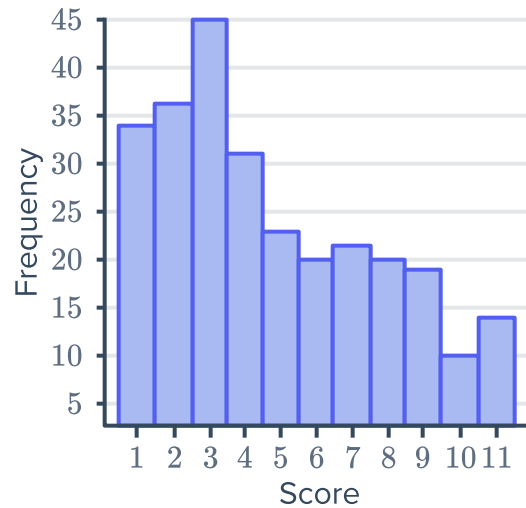
7 Consider the dot plot below:



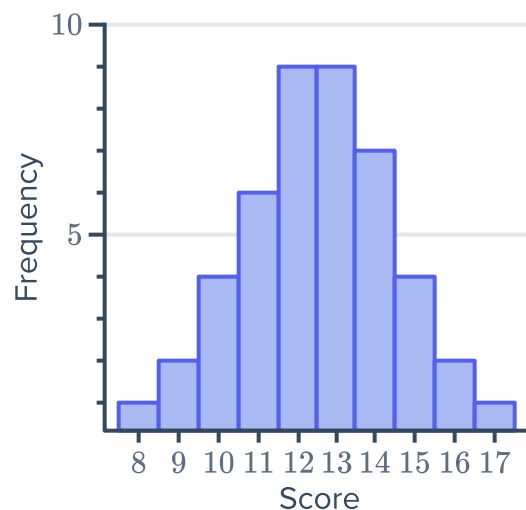
- a Are there any outliers?
- b Is there any clustering of data?
- c State the modal score(s).
- d Describe the shape of the distribution.

8 Consider the data shown in the histogram:

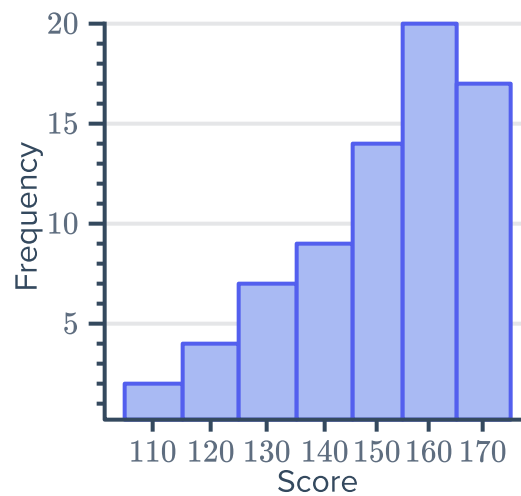
- a Are there any outliers?
- b Is there any clustering of data? If so, in what interval?
- c State the mode.
- d Describe the shape of the distribution.



9 Estimate the value of the mean of the following data set correct to one decimal place:



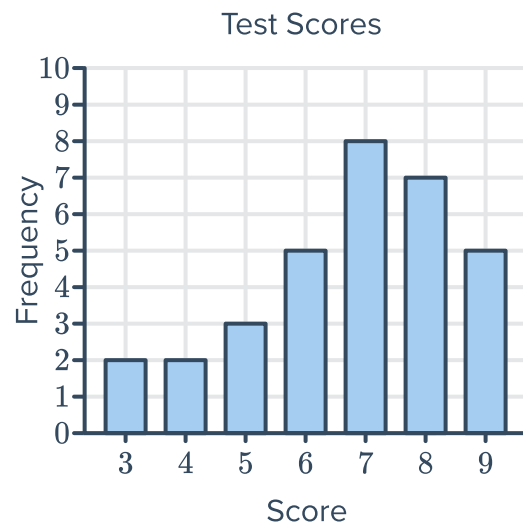
10 Consider the histogram representing students' heights in centimetres:



- a Does the histogram most likely represent grouped data or individual scores?
- b Estimate the value of the mean to one decimal place.
- c Describe the shape of the distribution.

11 Consider the given column graph:

- a Describe the shape of the distribution.
- b Find the following:
 - i Lower quartile
 - ii Upper quartile
- c Calculate the interquartile range.
- d Are there any outliers? If so, state the value.



12 Consider the dot plot given:

- a Describe the shape of the distribution.
- b Find the following:
 - i Lower quartile
 - ii Upper quartile
- c Calculate the interquartile range.
- d Are there any outliers? If so, state the value.



- 13 The stem and leaf plot below shows the ages of people to enter through the gates of a concert in the first 5 seconds:



- Find the median age.
- Find the difference between the least age and the median.
- Find the difference between the greatest age and the median.
- Calculate the mean age, correct to two decimal places.
- Is the data positively or negatively skewed?

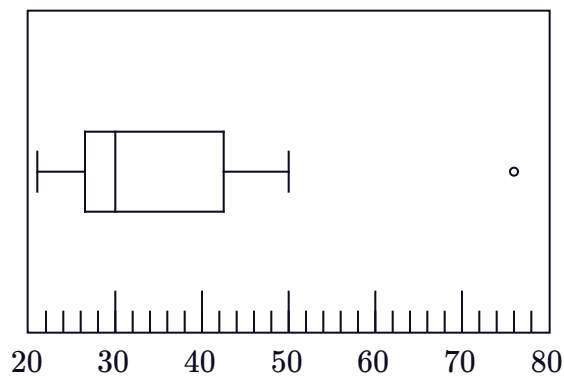
Leaf	
1	0 1 2 2 3 3 4 4 4 8 8 8
2	1 7
3	4 5 5
4	0
5	4

Key: 1|2 = 12 years old

- 14 $VO_2\text{Max}$ is a measure of how efficiently your body uses oxygen during exercise. The more physically fit you are, the greater your $VO_2\text{Max}$. Here are some people's results, listed in ascending order, when their $VO_2\text{Max}$ was measured:

21, 21, 23, 25, 26, 27, 28, 29, 29, 29, 30, 30, 32, 38, 38, 42, 43, 44, 48, 50, 76

- Determine the median $VO_2\text{Max}$.
- Determine the upper quartile value.
- Determine the lower quartile value.
- Consider the box plot for this data set and state whether the results are positively or negatively skewed.
- State the value of the outlier.
- An average untrained healthy person has a $VO_2\text{Max}$ between 30 and 40. What can we say about the exercise habits of the majority of this group of people?

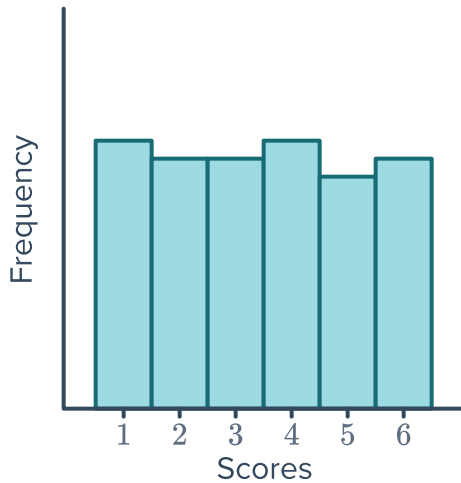


15 A die is rolled for a large number of trials and a number appearing is noted.

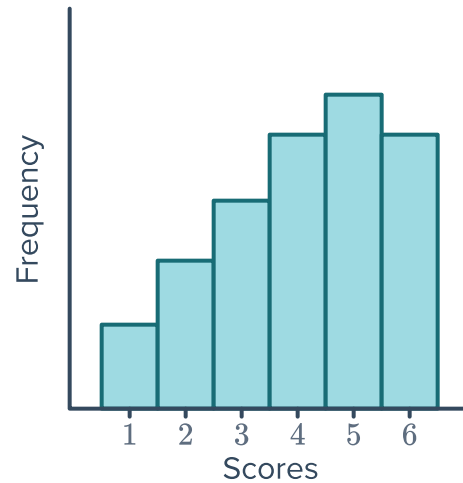


a Which histogram would you expect to match the data? Explain your answer.

A



B



b Describe the shape of the distribution of the data chosen above.

16 A pair of dice are rolled and the numbers appearing on the uppermost face are added to create a score.

a How many combinations would result in a score of:

i 4

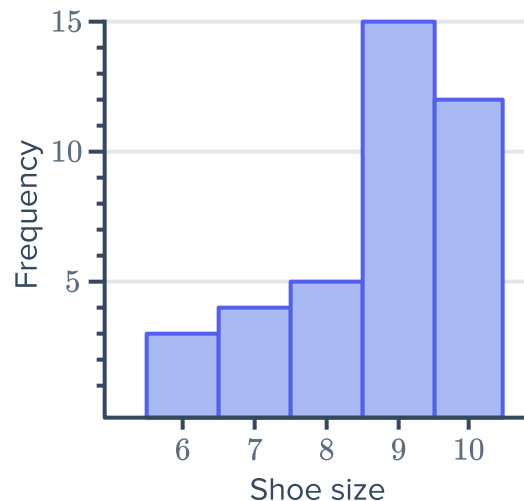
ii 7

iii 10

b A pair of dice are rolled for a large number of trials and the numbers appearing are added to create a score. Draw a sketch of a histogram you would expect to match the data.

17 The shoe sizes of all the students in a class were measured and the data was presented in a graph:

- a Are there any outliers?
- b Is there any clustering of data? If so, in what interval?
- c What is the modal shoe size?
- d Describe the shape of the distribution.

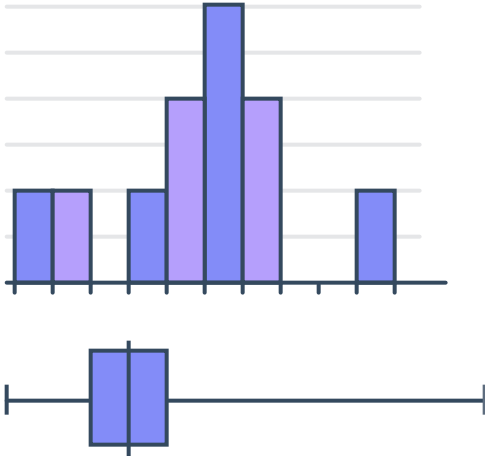




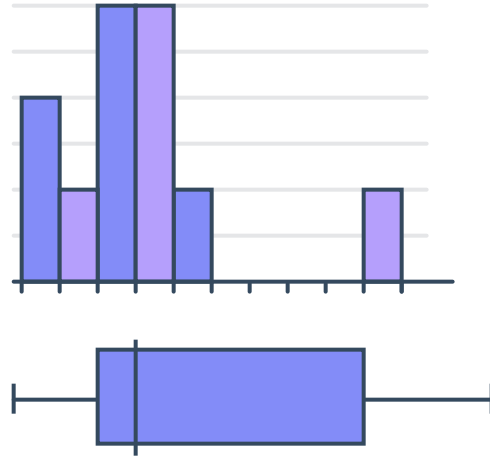
Histograms and box plots

18 Explain why the following pairs of histograms and box plots do not match:

a



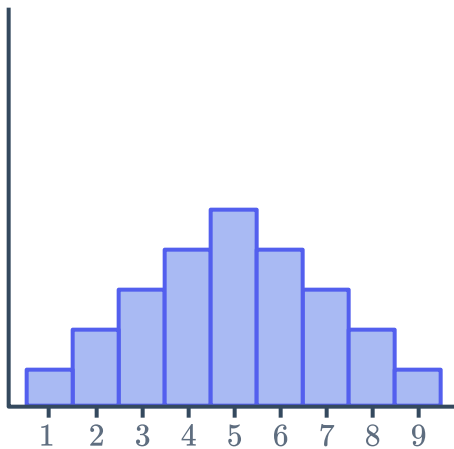
b



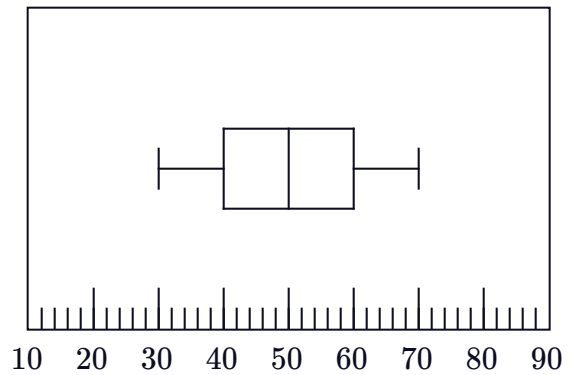
19 Match the histograms on the left to the corresponding box plots on the right:



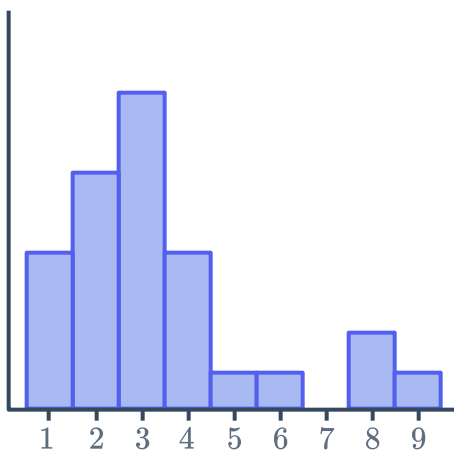
Histogram A



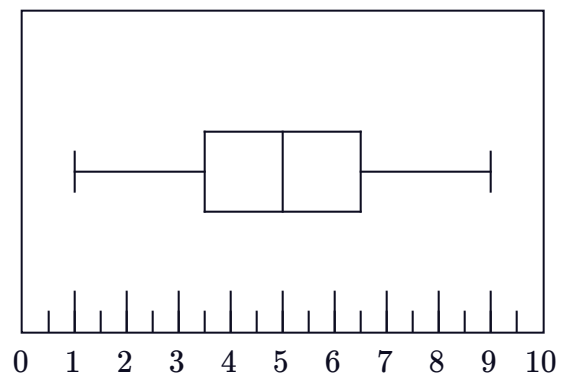
Box Plot 1



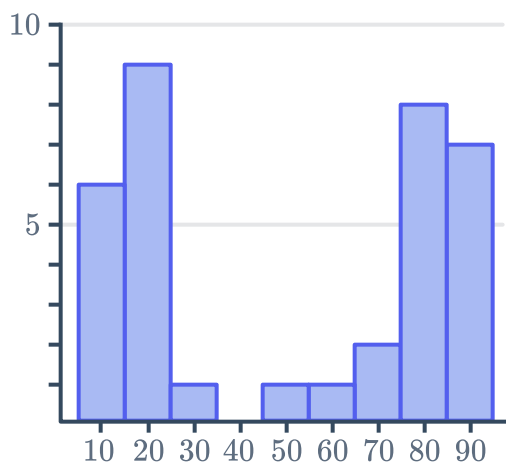
Histogram B



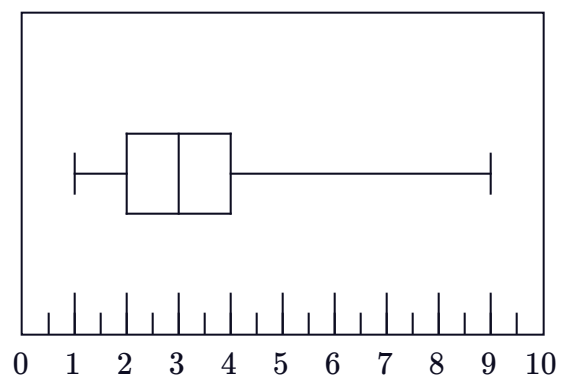
Box Plot 2



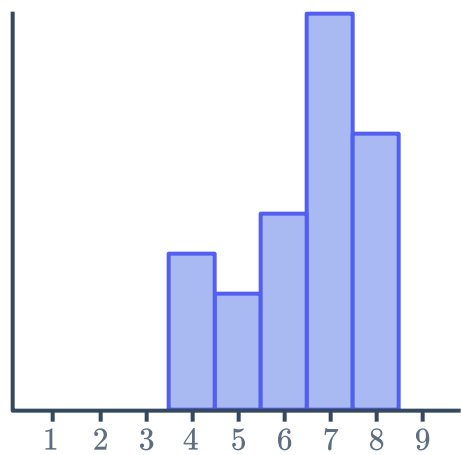
Histogram C



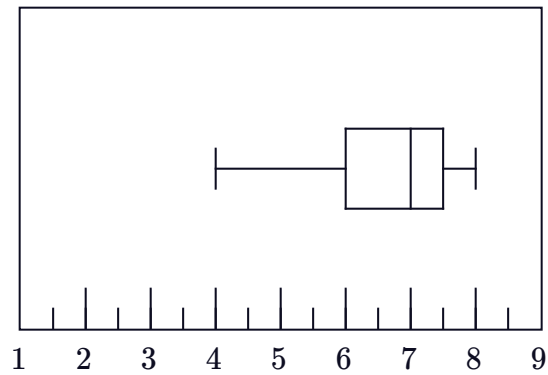
Box Plot 3



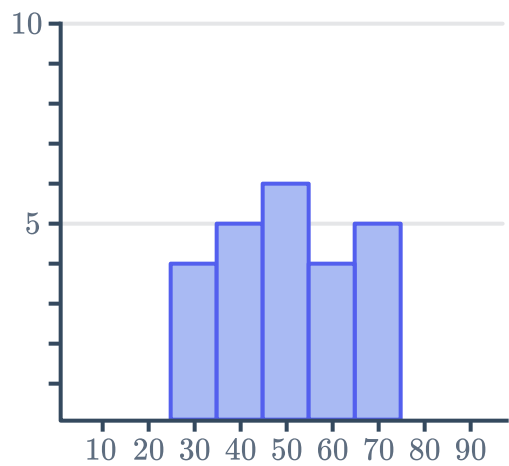
Histogram D



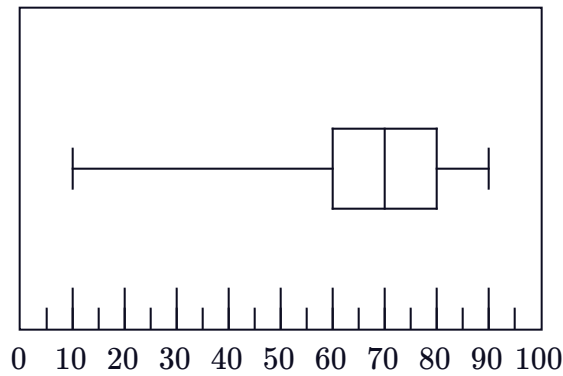
Box Plot 4



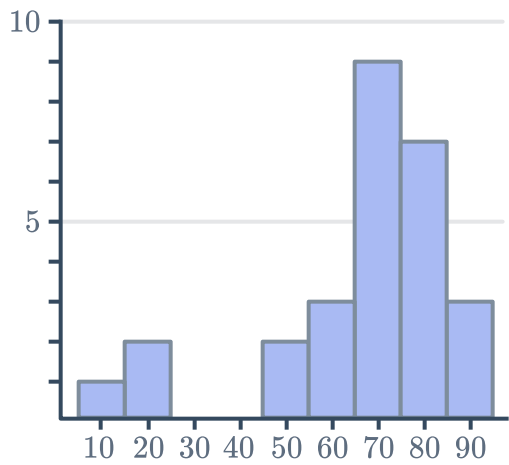
Histogram E



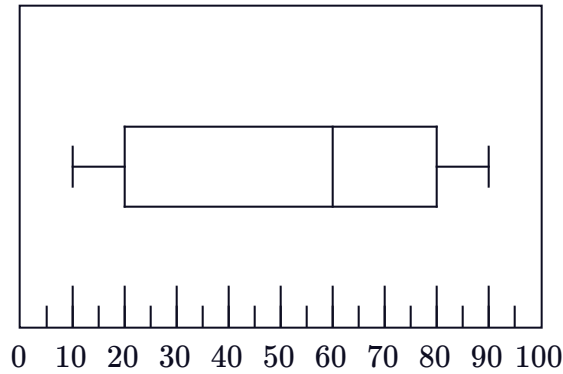
Box Plot 5



Histogram F



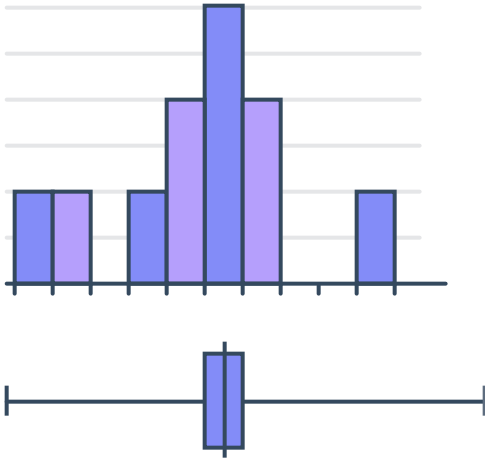
Box Plot 6



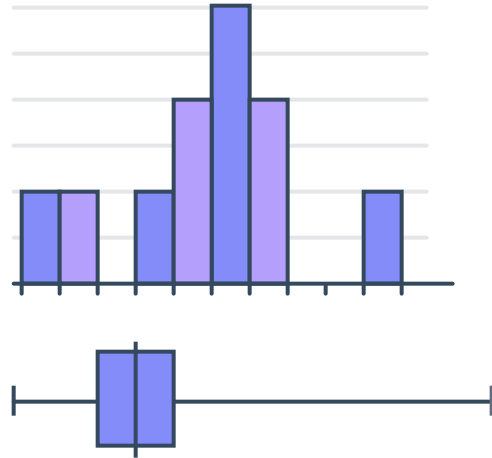
20 State whether the following pairs of histograms and box plots are correctly matched with respect to their shape:



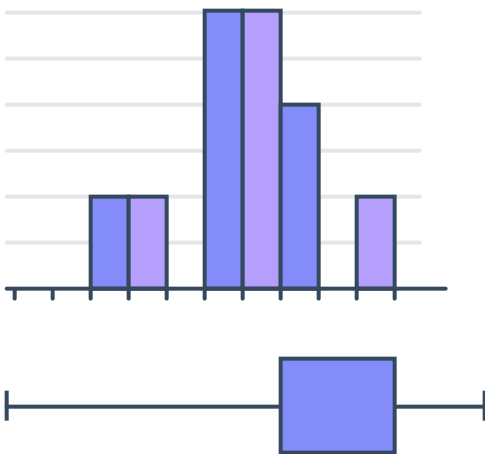
a



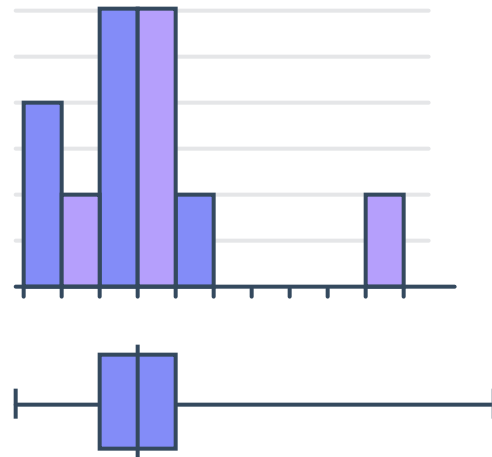
b



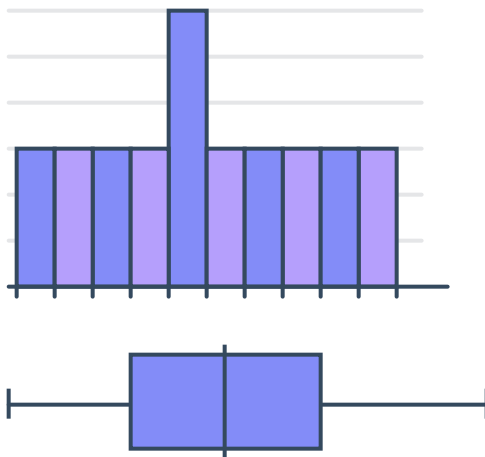
c



d



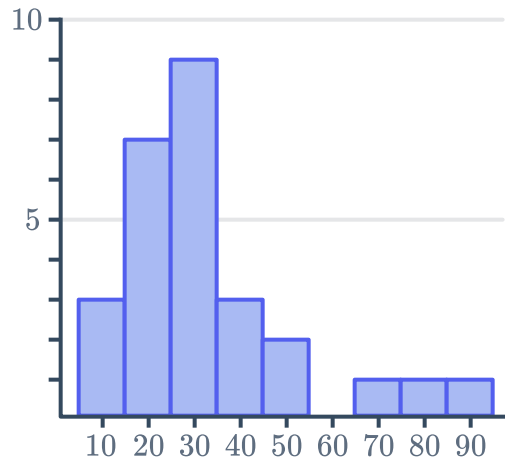
e



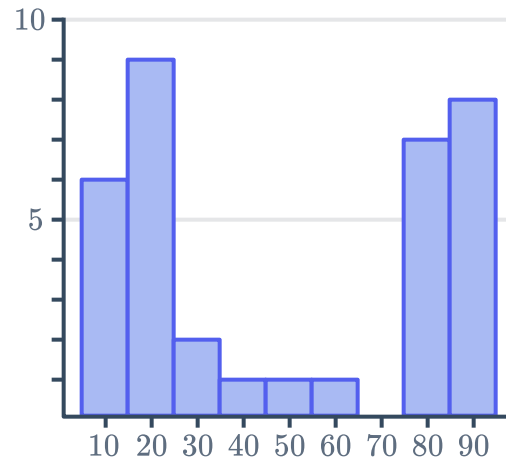
21 Construct a box plot for the following histograms



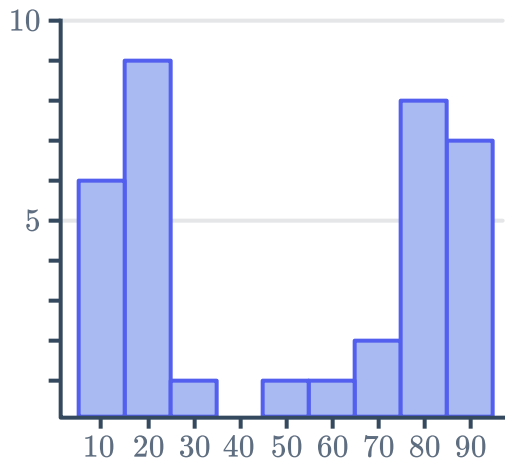
a



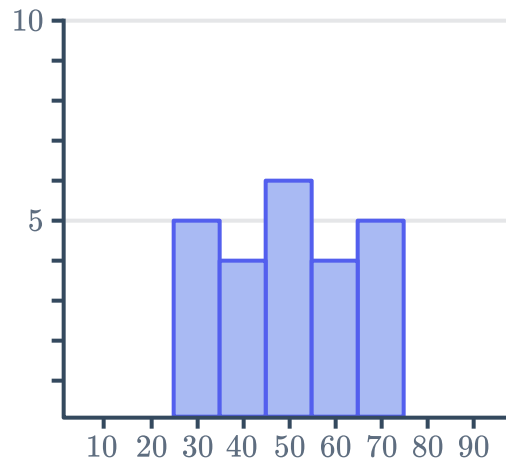
b



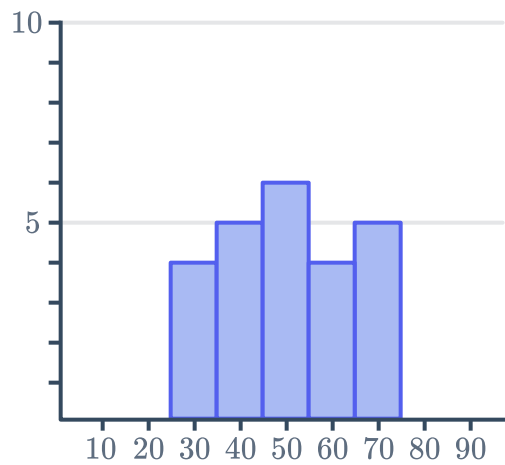
c



d



e



f

